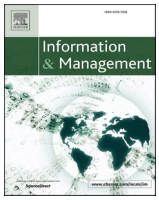
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Good becomes better? Temporarily acquired virtual badges and the signaling effects on online healthcare platforms

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ABSTRACT

Online healthcare platforms (OHPs) have become increasingly popular among physicians and patients. To facilitate patients' demand for physicians' services, some OHPs introduce non-financial, gamification designs such as temporarily acquired virtual badges. These badges are awarded periodically based on physicians' recent performance, subject to acquisition, retention, and loss. However, the impact of such strategies on patients is unclear and is understudied. We fill this gap by examining the signaling effects of temporarily acquired virtual badges on patients' subsequent participation in online consultations. We find that the acquisition and retention of such badges can improve patients' participation in online consultations. In addition, we identify factors that moderate these effects, including physicians' seniority and altruism. Our findings provide theoretical explanations and practical implications.

1. Introduction

Online healthcare platforms (OHPs) have been rapidly growing and embraced by an increasing number of patients and physicians in recent years, with 88 % of consumers and 83 % of physicians reporting having used or provided online medical services [1]. While OHPs eliminate time and distance barriers and address inequalities in healthcare access [2], it is difficult for patients to appropriately evaluate the quality of these medical services [3]. Therefore, it is critical for OHPs to assist patients in identifying high-quality online medical consultations and the associated physicians. To address this practical challenge, OHPs have employed virtual badges to recognize physicians based on their performance in medical consultations [4,5]. Such badges are typically shown on the physician's homepage permanently.

However, some argue that such permanent badges (that are valid permanently and cannot be revoked [6,7]) may not effectively reflect their recipients' performance, which may raise concerns about the ability of such badges to facilitate users' decision-making. For example, permanent badges are usually awarded based on the previous achievements and may not reflect the recipient's current performance, thus offering limited assistance in user decisions [8]. In addition, the acquisition of virtual badges can be manipulated, making them less reliable and thus hindering users' decision-making [9]. To address the shortcomings of permanent badges, some practitioners have recently suggested that the temporarily acquired virtual badges can be a potential solution [9]. In response to this practical call, our study investigates the impact of temporarily acquired virtual badges on OHPs, where physicians earn these badges based on their performance in the immediately preceding period. Such badges may undergo dynamic, periodic evaluations and status updates, including badge acquisition, retention, and loss.

The key differences between temporarily acquired and permanent badges are twofold: First, the evaluation approach for the two types of badges is different. Temporarily acquired badges are evaluated periodically based on physicians' performance in the preceding period. They can be revoked due to subpar performance or regained through improved performance in the next evaluation period. In contrast, permanent badges are assessed in a one-time evaluation based on the physicians' cumulative performance up to the point of evaluation. Second, temporarily acquired badges emphasize the current or the short-

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term status of acquiring, retaining, or losing badges within each evaluation period. Conversely, permanent badges represent an enduring status that stays with the physician after the award, unaffected by the physician's subsequent performance.

These fundamental differences between permanent and temporarily acquired badges can be better understood through signaling theory, which suggests that the credibility of a signal depends on its cost and observability [10]. Temporarily acquired badges are more costly than permanent badges, as the former requires physicians to continuously perform well in each short period (e.g., a month). In contrast, permanent badges do not require physicians' ongoing effort. Regarding signal observability, while permanent badges remain visible permanently and cannot be revoked after acquisition, temporarily acquired badges can be observed in three states: acquisition, retention, and loss, sending different observable signals to patients. These differences in signal cost and observability suggest that temporarily acquired badges may send more credible signals to patients, potentially enhancing their trust and helping them make informed consultation decisions [11]. Nevertheless, temporarily acquired badges may also have potential unintended effects. For example, the temporary nature of these badges might not convey long-term prestige and recognition to physicians as permanent badges do, which may affect patients' participation in online consultations and physicians' motivation to engage in OHPs [12].

Despite the importance of temporarily acquired badges to patients, physicians, and OHPs, their effectiveness displayed through different statuses of acquisition, retention, and loss remains unclear. Filling this research gap is important for several reasons. First, patients rely on credible signals (such as virtual badges) to build trust in physicians and assess the performance of their services, alleviating information asymmetry between physicians and patients [13]. The novel design of temporarily acquired badges enhances the signaling by detailing the processes of badge acquisition, retention, and loss, providing patients with more frequent and credible signals than permanent badges. Therefore, studying the effects of temporarily acquired badges contributes to the literature on signaling theory within the context of OHPs. Second, understanding the effects of temporarily acquired badges can benefit all major stakeholders. For patients, it may influence satisfaction and engagement by providing more up-to-date information for decision making. For physicians, it may affect their motivation for engagement in OHPs [14]. For platforms, it introduces a new incentive strategy with potentially positive effects on user engagement [15]. Overall, these impacts can significantly affect the major stakeholders in online healthcare and the sustainable development of OHPs [16,17].

Therefore, our study aims to investigate whether and how dynamic changes of temporarily acquired virtual badges (including acquisition, retention, and loss) affect patients' subsequent participation in online consultations (i.e., consultation volume). Therefore, we raise our first research question: *Does the dynamic nature (*i.e.*, acquisition, retention, loss) of virtual badges have significant impacts on patients' subsequent participation in online consultations (*i.e.*, consultation volume)? If so, how?*

In practice, some industry experts suggest that OHPs need to consider individuals' characteristics when designing platform strategies (e.g., virtual badges) because these characteristics may influence the effectiveness of such platform strategies [18]. In response to the practical needs, we examine the heterogeneous effects of physicians' characteristics, specifically focusing on seniority and altruism. Previous research indicates that physicians' seniority and altruism represent their medical competence and benevolence, respectively, which are critical components of trust [19,20]. These characteristics can enhance patients' trust in physicians and thus moderate the effect of virtual badges on patients' subsequent consultation decisions [21,22]. Importantly, these two physician characteristics may exert distinct moderating influences on the effect of temporarily acquired badges on patients' participation in online medical consultations by conveying signals similar to or different from those of the badges. However, the specific roles of physicians' seniority and altruism in the signaling effects of temporarily acquired badges have not been empirically studied in the context of OHPs. To study the heterogeneous effects of temporarily acquired virtual badges for different physicians, we propose our second research question: *Do physicians' key characteristics (*i.e.*, seniority and altruism) moderate the impact of virtual badges on patients' subsequent participation in online consultations (*e.g.*, consultation volume)? If so, how?*

To empirically test our hypotheses, we collect data from a renowned OHP in China and quantify the effect of temporarily acquired virtual badges on patients' subsequent participation in online consultations (e.g., consultation volume of physicians). Additionally, we investigate and measure the moderating effect of physicians' seniority and altruism on the main relationship between temporarily acquired virtual badges and patients' subsequent participation in online consultations. Given that when OHPs award badges and the selection of physicians is not random, we introduce three identification strategies to address potential endogeneity concerns and perform various robustness checks.

Our analyses yield a range of robust and consistent results. First, we find that physicians acquiring and retaining virtual badges positively impact patients' subsequent participation in online consultations, with the effect of retaining badges being greater than that of acquiring them. However, the impact of losing badges is insignificant. We explain our findings based on signaling theory. Specifically, the acquisition and retention of virtual badges signal physicians' performance and competence, which enhances patients' trust and facilitates their participation. Compared to acquiring badges, the cost and observability of retaining them are higher, which fosters greater trust in the virtual badges among patients, thus leading to a more significant positive impact on consultation volume.

Moreover, we identify the heterogeneous moderating effects of physicians' characteristics. Specifically, physicians' seniority negatively moderates the impact of virtual badges on patients' subsequent participation in online consultations. This suggests that virtual badges are more effective in increasing consultation volumes for junior physicians by compensating for the initial lack of patients' trust and thereby promoting consultation decisions. Conversely, senior physicians with higher patients' trust experience a reduced influence from the virtual badges. Furthermore, physicians' altruism positively moderates the impact of virtual badges on consultation volume. This implies that altruism, as a signal of benevolence, provides additional trust cues for patients. This reassurance can enhance patients' confidence in decision making, thereby increasing consultation volumes.

Our findings provide practical implications for using and designing non-financial, gamification designs (i.e., temporarily acquired virtual badges) on OHPs and other online platforms. First, our study provides insight into the effectiveness of virtual badges on online platforms by showing the positive role of temporarily acquired badges in the context of OHPs. Specifically, physicians acquiring and retaining temporarily acquired badges can increase patients' participation in online consultations, with retaining badges having a greater impact. Additionally, we find that badge loss does not immediately yield the expected negative impact but takes time to manifest. Second, our study provides practical implications for OHP managers, stating that OHPs can further leverage the utility of temporarily acquired badges by enhancing their visibility, such as providing information on the cost and effort required for physicians to acquire the badge. Similarly, these badge designs can be extended to other two-sided online platforms like e-commerce and knowledge-sharing platforms. Third, OHP managers can tailor the design of virtual badges to maximize their effectiveness based on different physician characteristics (e.g., seniority and altruism). For example, managers of OHPs should focus more on non-senior physicians when designing and optimizing temporarily acquired virtual badges.

Our study contributes to the literature on OHPs, platform strategies, and signaling theory. First, our study contributes to the literature on patients' engagement in OHPs by examining the signaling effects of temporarily acquired virtual badges on patients' subsequent participation in online consultations. Furthermore, we emphasize how

temporarily acquired virtual badges generate differential signaling effects among physicians with different characteristics. Second, our study enriches the research on the effectiveness and design of virtual badges on online platforms by investigating the dynamics of temporarily acquired virtual badges. In contrast, previous research has primarily focused on the effects of permanent badges (e.g., one-time achievement badges [23]). Third, we extend signaling theory to the online healthcare context, highlighting the unique impact of recency–attribute signals based on recipients' recent performance. Unlike accumulation–attribute signals derived from recipients' accumulated performance, our research examines the effects of recency–attribute signals based on recent performance on patients' outcomes, thus providing a theoretical foundation for understanding comprehensive signaling effects.

2. Literature review

In this section, we present three literature streams related to our research: (1) online healthcare platforms, (2) the impact of platform strategies on consumer behaviors, and (3) signaling effects on OHPs. We review the relevant literature and highlight the differences between this research and past literature.

2.1. Online healthcare platforms

The online engagement of physicians and patients on OHPs is crucial for both patients' health outcomes and the sustainability of OHPs. Physicians' engagement in OHPs and the provision of online medical consultations can facilitate collaborative physician–patient care and support patients' self-management, thereby improving patients' well-being and the physician–patient relationship [16], as well as increasing patients' review valence [24]. Patients' engagement in OHPs can empower them to better manage their health by connecting with others [25]. Such engagement not only increases patients' perceived usefulness of health information and improves satisfaction [26,27] but also improves self-care efficacy and perceived health outcomes [28].

In addition, several studies have further explored the factors influencing the online engagement of patients and physicians [16,28,29]. For patients, certain platform features (e.g., reputation systems) can enhance their trust in physicians, thereby facilitating their online engagement. For example, displaying the online reputation of physicians and hospitals on OHPs can improve patients' trust in physicians, encouraging patients to choose online consultations with physicians [30]. The introduction of a Q&A feature on OHPs can help patients distinguish between high-ability physicians and low-ability physicians, thereby influencing patients' recommendations for physicians [31]. For physicians, relevant studies have discussed the impact of incentive strategies on their online engagement. For example, introductory financial incentives provided by OHPs can increase the number of contributions by new physicians during the campaign period but decrease in the long term, and such incentives are detrimental to the quality of physicians' contributions [32].

The research to date has explored the effects of physicians' and patients' engagement in OHPs, as well as the impact of platform features within OHPs or the incentives provided by OHPs on their online participation. However, limited studies have considered the effect of non-financial, gamification design (such as virtual badges) on patients' participation in online consultations with physicians. With the increasing prevalence of virtual badges on OHPs, it is essential to elucidate their impact on patients' participation in online consultations, which may affect the benefits and development of physicians and OHPs. The findings can help OHP managers and healthcare practitioners understand the effectiveness of virtual badges, thereby providing practical insights to facilitate patient demand in the online healthcare context.

2.2. Impact of platform strategies on consumer behaviors

A body of research has investigated strategies used on online platforms, such as the establishment of reputation systems [30,33], feedback systems (e.g., online reviews) [34,35], and recommendation systems [36] and their impact on consumer behavior and decision-making. For example, a high level of seller reputation could help consumers eliminate uncertainty, thereby assisting sellers in attracting more consumers [33]. Online reviews that better characterize patient's perceived service quality could significantly increase the possibility of potential patients' choices [35]. Providing free Q&A services allows physicians to convey credible quality information to patients, thereby increasing patients' recommendations for physicians [31].

In addition, there is a stream of information systems literature about virtual badges, a gamified design that is awarded by online platforms based on users' achievements [37,38]. Previous literature has explored the effect of virtual badges on users' behaviors in online platforms such as e-commerce and review platforms [39–42]. For example, in e-commerce, platforms like Taobao have introduced Gold Medal Seller virtual badges to signify seller quality, reducing consumer uncertainty and facilitating their purchase decisions [11]. In the sharing economy context, the gamification design of the "Superhost" badge to hosts can increase the number of consumer reviews and improve their review ratings [40]. On an online review platform, the "Elite" badge on Yelp can enhance the perceived value of reviews, attracting more useful votes from users [42]. In the context of online healthcare, Haodf.com awards the "Annual Physician" badges to physicians, which can attract more patients' appointments and review volume [41].

In sum, previous literature has primarily explored the impact of platform strategies, such as reputation systems and virtual badges, on consumer behavior. We present the positioning of our study in the related literature in Table B1 in E-Companion B. Furthermore, we have clarified the differences between our study and previous studies and highlighted our contributions as follows. First, the virtual badge design in prior literature has mainly been conducted in e-commerce and other online environments, with limited research in the context of OHPs. Our research extends this stream of literature to the online healthcare context and studies the effectiveness of virtual badges, particularly the temporarily acquired badges with different states, on patients' participation in OHPs.

Second, past studies have primarily studied the impact of the evaluation of accumulative performance over a period on consumer's subsequent behavior [41]. In contrast, our study focuses on the effectiveness of temporary evaluation (e.g., virtual badge), which is a timely assessment and feedback that evaluates user performance in each short period (e.g., a month) on consumers' subsequent engagement. Our research on virtual badges based on temporary evaluation sheds light on the efficacy of this evaluation approach.

Third, most studies on virtual badges assume they are, once earned, permanently possessed, with scarce consideration of badge acquisition, retention, and loss. Only a handful of studies have delineated the changes in the badge states (e.g., [11]). However, our study differs from those studies in several major aspects: (1) We study the effects of virtual badges in the context of OHPs, while previous studies have primarily focused on the context of e-commerce platforms. The type and effectiveness of virtual badges may differ in different research contexts. We elaborate on it in the next two aspects. (2) The badges in our study are based on temporary evaluation, whereas badges studied in prior research, such as Cheng et al. [11], are based on merchants' cumulative performance. (3) We find that physicians losing badges does not immediately affect patients' subsequent consultation behavior. Instead, such a badge-losing state takes effect over time, which contrasts with findings in the literature that losing badges immediately reduces merchants' sales.

Overall, our research contributes to the literature on platform strategies, especially on the strategy of offering temporarily acquired virtual

badges, by investigating the impact of dynamic changes of temporarily acquired badges (including badge acquisition, retention, and loss) on patients' subsequent engagement in online consultations. This is an important question for platform managers, as some industry experts point out that the temporarily acquired badge design [9] can address issues such as manipulation due to cumulative information [11] or the outdated information of service providers [43], which may affect consumers' trust in virtual badges and their subsequent participation behavior. Furthermore, our study aims to provide practical insights for OHP managers, helping them understand how temporarily acquired virtual badges might affect patients and design different implementation strategies for different physicians accordingly.

2.3. Signaling effects on OHPs

Signaling theory posits that in situations of information asymmetry, where one party possesses more information than the other, effective signals can be utilized to bridge this gap and alleviate uncertainty [10]. Therefore, signaling mechanisms are widely used to explain the formation of users' attitudes and trust toward online platforms and their subsequent behavioral changes. For example, merchants on e-commerce platforms can increase their sales by acquiring reputation badges, as such badges convey quality signals of the merchants and their products to potential consumers, thereby enhancing consumer trust and facilitating their payment decisions [11]. In online healthcare, physicians typically know more about patients' conditions and treatment options than the patients themselves [35]. Conversely, patients generally lack the ability to discern the quality of medical advice provided by physicians. This phenomenon results in information asymmetry between physicians and patients, hindering the decision-making process for patients when choosing physicians for consultations.

To improve patients' demand for physicians' services, OHPs provide various signal mechanisms to help patients identify high-ability physicians and explore the effects of signals [22,44,45]. For example, Chen et al. [44] investigated the effectiveness of different signals (i.e., owned and earned signals) on patient demand. Specifically, owned signals reflect the credentials or professional status that physicians possess, while earned signals represent the aggregated evaluation of physicians' online medical activities on OHPs, such as online reviews posted by patients. Both types of signals can influence patient demand. Zhou et al. [22] categorized physicians' signals into description signals and demonstration signals, where the former indicates professional qualifications and consultation style, and the latter indicates physicians' responsiveness to patients' questions. These signals derive from personal profiles or accumulated service records displayed on physicians' homepages, which can influence potential patients' evaluations of physicians' quality and their demand for physicians' services.

Overall, existing research on signaling effects on OHPs focuses on accumulation–attribute signals generated from physicians' cumulative performance (e.g., an annual recognition for outstanding physicians [8, 41]) or historical information. However, some studies have found that such accumulation–attribute signals may lose credibility over time and are subject to manipulation risks [11,46–48]. In contrast, the temporarily acquired virtual badges examined in our study offer a potential solution to these issues. Specifically, such virtual badges evaluate only the recent medical activities of physicians and convey recency–attribute signals rather than accumulation–attribute signals. This up-to-date assessment of recency–attribute signals captures changes in physicians' capabilities [43], which may enhance the credibility of signals. Despite these potential benefits, research on the effects of recency–attribute signals is limited. Therefore, we contribute to the literature on signaling effects by investigating the impact of recency–attribute signals conveyed through the dynamic changes of temporarily acquired virtual badges.

In addition, we explore the moderating role of physicians' key characteristics (i.e., seniority and altruism) to better understand the heterogeneity in the signaling effect of temporarily acquired virtual badges. Prior research indicates that the impact of signaling on individuals' online engagement may vary depending on the signal sender's level of seniority and altruism. For example, patients perceive that senior physicians may possess higher expertise and richer experience, thereby establishing more trust in the quality of physicians' medical services, which results in more consultation appointments [49]. In addition, patients may perceive physicians who exhibit altruism as being more caring about patients' needs, leading to more online consultations and greater satisfaction toward physicians' services [50,51]. Drawing on the literature, we choose physician seniority and altruism as moderating variables and suggest that they may moderate the extent to which patients trust physicians, thereby moderating the signaling effect of temporarily acquired badges on patients' participation in consultations. Our study contributes to the literature on the moderating role of physician seniority and altruism on OHPs.

3. Hypotheses development

In this section, we provide a theoretical foundation from the signaling and trust perspectives for understanding how the dynamics of temporarily acquired virtual badges affect consultation volume, as well as the heterogeneous impacts of different physicians' characteristics. We begin by introducing signaling theory and discussing the impact of virtual badges as a credible signal of physicians' performance and competence on patients' trust and their subsequent consultation decisions. Then, given the dynamic nature of virtual badges, we develop hypotheses for the signaling effects of different badge statuses (i.e., acquisition, retention, and loss) on consultation volume in §3.1–3.3. At last, we discuss the role of physicians' characteristics (i.e., seniority and altruism) as additional signals in establishing patients' trust in physicians and propose hypotheses on the heterogeneous effects of virtual badges in §3.4–3.5.

3.1. Signaling impact of acquiring virtual badges on patients' subsequent participation in online consultations

To understand the impact of virtual badges, we leverage a theoretical framework of signaling theory [10]. The premise of signaling theory is the existence of information asymmetry and the potential for divergence or conflict of interest between the signaler and receiver [49]. In online healthcare, physicians generally know more about patients' conditions and needs, but patients have limited insights into the qualities of physicians and their services [35], indicating an apparent information asymmetry between physicians and patients. To address this problem, healthcare providers can help patients make consultation decisions by using effective signals that can differentiate between physicians with different performance [45].

A virtual badge is a form of recognition granted by online platforms (such as question-and-answer platforms and online review communities) to users who have made certain contributions [37,38]. Such badges recognize the recipients' previous achievements in certain activities [52]. In our context, virtual badges are non-financial recognitions granted by OHPs based on physicians' previous comprehensive performance in online medical activities, assessed through key indicators such as response speed, medical service outcomes, and physician service attitudes. Such badge evaluation mechanism suggests that it would be more costly for physicians with low previous performance (low quality) on OHPs to acquire virtual badges than those with high previous performance (high quality), as the former would need to invest more effort and time to meet the badge evaluation criteria.

Especially for the temporarily acquired virtual badges, which are assessed monthly, it can be more costly for physicians with subpar performance to improve their performance in the short term to meet the badge evaluation criteria. According to signaling theory, signals are more credible when they are costly to convey, as senders with lower

performance are less likely to bear such costs to mimic the signals sent by more competent individuals [53]. Consequently, temporarily acquired badges can serve as a credible signal to help patients discern physicians' performance in online medical services.

In addition, prior research indicates that competence is an important component in building trust, which plays a critical role in the consumer decision-making process [20]. Similarly, in online healthcare, physicians' competence may play a foundational part in building the trust of patients [44]. Along this line, we expect that acquiring virtual badges can differentiate well-performing physicians from those with not-so-well performance, thereby signaling to patients the high performance and competence of the badge recipients. This signal is expected to increase patients' trust in physicians and subsequently facilitate patients' consultation decisions, as reflected in the volume of consultations. Thus, we propose our first hypothesis as follows:

Hypothesis 1. (H1): *Physicians' acquisition of virtual badges may exert a positive signaling impact on patients and increase patients' subsequent participation in online consultations (*e.g.*, increase consultation volume).*

3.2. Signaling impact of retaining virtual badges on patients' subsequent participation in online consultations

Typically, OHPs award virtual badges monthly based on physicians' previous performance and display the information about whether a physician has received or maintained a virtual badge on the physician's homepage. Therefore, given the dynamic nature of virtual badges, beyond the acquisition, the retention of virtual badges (i.e., acquiring badges twice in a row) may exert a different signaling effect on patients' subsequent participation in online consultations.

Previous research has shown that the credibility of signals increases when they exhibit higher cost and observability [10]. Regarding signal costs, badge retention potentially incurs higher costs than mere acquisition. As retention includes both stages (i.e., first acquiring and then retaining badges), physicians must consistently invest time and effort to provide high-quality medical services [54] during the two stages, compared to only one stage. This requirement suggests that, apart from the initial costs in terms of resources or efforts needed to acquire badges, retaining badges may signify added costs for physicians in ensuring consistent services. Regarding signal observability, the retained badges on physicians' homepages make it easier for patients to obtain confirmation about physicians' consistent performance. The extended display of these badges implies greater observability of signals. Therefore, in our context, physicians' retaining virtual badges may serve as a more credible signal than merely acquiring virtual badges.

In addition, previous research has shown a positive correlation between signal credibility and receivers' trust in the sender, and the latter may further influence the receivers' subsequent decision making [22]. To summarize, as signals reveal higher cost and observability, the signals may exhibit higher credibility, which may further enhance the signal receivers' trust. Therefore, based on the literature, we propose that physicians' retention of virtual badges may enhance patients' trust by demonstrating physicians' consistency and reliability of superior performance. This enhanced trust may, in turn, lead to more patients' participation in consultations, amplifying the effects of the initial badge acquisition. Put together, we propose that:

Hypothesis 2. (H2): *(a) Physicians' retention of virtual badges may exert a positive signaling impact on patients and their subsequent participation in online consultations (*e.g.*, consultation volume), (b) and the magnitude of this effect is larger than that for physicians' acquisition of virtual badges.*

3.3. Signaling impact of losing virtual badges on patients' subsequent participation in online consultations

As previously mentioned, the OHP's evaluation of whether physicians can acquire a virtual badge is dynamic. If physicians with virtual badges fail to meet the platform's evaluation criteria in a given month, then their badges are withdrawn by the platform during that month. If, as we previously hypothesized, acquiring and retaining virtual badges are positive signals, then it is reasonable to posit that losing virtual badges should be a negative signal. Specifically, the loss of badges may reduce the signal cost because if a physician fails to retain a badge, patients can no longer perceive the physician's performance and competence in online medical services through the badge. Additionally, the loss of badges may hinder the observability of the signal because the badge is no longer available for patients to observe.

In addition, related research in e-commerce shows that losing a badge can negatively impact sellers' sales, as the loss of badges causes consumers to reduce their perception of sellers' quality [11]. Similarly, in online healthcare, physicians losing virtual badges may reduce patients' trust in physicians as patients' perception of physicians' performance and competence may be negatively affected. Therefore, it is difficult for physicians without virtual badges to stand out from other physicians, reducing the likelihood of patients' choice of physicians. Therefore, we hypothesize:

Hypothesis 3. (H3): *Losing virtual badges may have a negative signaling impact on patients and their subsequent participation in online consultations (*e.g.*, consultation volume).*

3.4. The moderating role of physicians' seniority

Prior literature suggests that the impacts of online signals may vary among different physicians [4]. To explore the heterogeneity in the signaling effects of virtual badges across different physicians' characteristics, we discuss the moderating roles of two features that can influence the establishment of patients' trust in physicians (i.e., seniority and altruism). In addition, prior research suggests that seniority and altruism are two key characteristics of physicians and thus may moderate the signaling effects of virtual badges [21,22]. We first consider physicians' seniority as a moderating factor.

In China, physicians' seniority is typically represented by their clinical titles, a uniform rank system for evaluating physicians' professional qualifications [14]. Consequently, physicians with higher seniority are presumed to possess greater professional capability and experience [55]. Similar to virtual badges, seniority can signal physicians' performance and provide clues about their competence, which can influence patients' trust in physicians and affect their decision to consult [14,56,57]. Therefore, the similar nature of the signals conveyed by physicians' seniority and virtual badges may suggest that seniority might serve as a substitute for virtual badges in enhancing patients' trust and facilitating their consultation decisions.

On the one hand, high clinical titles require physicians to undergo rigorous training and assessment [44], which makes patients' trust in senior physicians sufficiently high. Therefore, we expect that having virtual badges provides limited help in further enhancing patients' trust in senior physicians. On the other hand, for junior physicians, having virtual badges can compensate for the lack of patients' trust in physician competence through seniority, thus enhancing patients' trust in physicians and increasing the consultation volume. Put together, we hypothesize that:

Hypothesis 4. (H4). *Physicians' seniority negatively moderates the impact of physicians' (a) acquiring, (b) retaining virtual badges on the consultation volume, while positively moderates the impact of physicians' (c) losing virtual badges on the consultation volume.*

3.5. The moderating role of physicians' altruism

Previous research on trust has indicated that, in addition to competence, benevolence is another component of trust [19]. Benevolence represents an ethical trait that refers to a person's motivation to act in order to help others rather than to serve themselves [58]. In the medical

context, similar to benevolence, altruism with a sense of service and self-sacrifice is considered an essential component of medical professionalism and one of the traditional values of doctoring [59,60]. Research has suggested that physicians with greater altruism have higher ideals for medical services than treating medicine merely as a profession [21]. Along these lines, we expect physicians with high levels of altruism to be more likely to be perceived as benevolent.

On our focus platform, physicians' altruism can be seen by patients through the pro bono identity displayed on the physician's homepage. This identity is a digital symbol granted by the platform to physicians who have engaged in a certain number of voluntary activities on OHPs, such as providing free consultations. Thus, altruism (through the pro bono identity) can serve as a signal of benevolence, providing patients with additional cues of trust. This, in turn, may make patients feel trust and assured about the physician and enhance patients' demand for the physician.

Previous research has shown that benevolence can synergize with other dimensions of trust (e.g., competence) to facilitate users' online decision-making behaviors [20,61]. For instance, consumers are more likely to trust merchants when they perceive merchants as benevolent (concerned about consumer interests) in addition to competent (showing good performance in meeting consumer needs) [61]. Along the same line, we propose that physicians' altruism can provide a signal of benevolence to patients, which may complement the signal of physicians' competence conveyed by virtual badges. Specifically, benevolence enhances the individuals' trust and emotional connection associated with the competence signal [62], which amplifies the badge's effect on consultation volume beyond its standalone impact. For example, badges alone signal "this physician can help me," while the physician's altruism signals "this physician wants to help me." Together, their combination produces a more compelling message: "This physician can and wants to help me." In other words, physicians' altruism may positively moderate the effect of virtual badges on increasing the consultation volume. Based on this discussion, we formally propose the following hypothesis:

Hypothesis 5. (H5). *Physicians' altruism positively moderates the impact of physicians' (a) acquiring, (b) retaining virtual badges on the consultation volume, while negatively moderates the impact of physicians' (c) losing virtual badges on the consultation volume.*

4. Research context and data description

4.1. Research context

Our research context is the Yilu app, one of the pioneer OHPs in China. As of March 2022, the number of monthly active users on Yilu has exceeded 3.2 million, and it ranks first among OHPs in the online healthcare industry [63]. We choose Yilu as our research context because the app is representative of OHPs and has a large user base. Moreover, the app offers the unique feature of temporarily acquired virtual badges for physicians. On this platform, physicians can provide registered patients with medical services such as online consultation, appointment booking, prescription, and medication delivery. After the consultation service, patients can evaluate the physicians and the service provided, and the evaluation is displayed on the physician's homepage. We present a screenshot of a representative physician's homepage in Figure A1 in E-Companion A.

Recently, Yilu has introduced temporarily acquired virtual badges to display comprehensive performance information about physicians' online medical activities, aiming to alleviate information asymmetry between physicians and patients, thereby aiding patients in selecting physicians for consultations. The platform assesses and determines physicians' eligibility for the badge for the current month based on physicians' comprehensive performance (e.g., response speed, the outcome of medical services, and the physicians' service attitudes) during the past month. The platform then ranks all physicians within each department based on their comprehensive performance and selects a maximum of the top twenty physicians as badge recipients for that month. These badges are dynamically updated each month, meaning physicians may acquire, retain, or lose badges in the current month relative to the previous month. The temporarily acquired badges are displayed on the recipient's homepage and the platform's recommendation list, which are easily observable by patients. Note that both physicians who receive badges and physicians who do not receive badges remain on the recommendation list throughout our study period. Such consistent exposure across both groups minimizes the potential confounding effect of the recommendation list on the impact of virtual badges. This ensures that the recommendation list does not confound the badge effect. More importantly, patients can click on the badge icon to access detailed information about the badge evaluation criteria, thereby enhancing the transparency of the badge selection process.

Understanding the role of temporarily acquired virtual badges has important implications for online healthcare stakeholders. For patients, a credible signal can promote their trust in physicians and OHPs, thereby assisting them in selecting appropriate physicians for their consultations. For platforms, understanding the signaling effects of temporarily acquired virtual badges can help them implement and design effective virtual badges to facilitate patient demand for the physician's service, thus ensuring the sustainable development of the platform.

4.2. Data description

We collected a six-month panel dataset containing physicians' homepage information (e.g., department, title, number of consultation services) and patients' evaluations (e.g., review tags, textual reviews) from March 2022 to August 2022. Our analysis is at the monthly level for each physician, allowing us to capture the changes in their online consultation volume after receiving temporarily acquired virtual badges. We present summary statistics of the key variables and their definitions in Table 1.

Our dataset contains information about the personal attributes, online behaviors, and patients' evaluations of 6481 physicians. As mentioned earlier, our dependent variable is physicians' online consultation volume. As shown in Table 1, each physician on the platform conducts an average of 21 (logarithm value is 1.199) consultation services per month. Regarding the temporarily acquired virtual badges, the average probability of a physician having a virtual badge in a given month is about 2 %, suggesting a potential practical effect of badges that differentiates recipients from other physicians. In addition, the average probability of a physician not having a badge in the past month but acquiring one in the current month is 0.4 %. The average probability of a physician retaining a badge from the past month to the current month is 1.3 %. Meanwhile, the average probability of a physician having a badge in the past month but losing it in the current month is 0.5 %. Among physicians who have badges in the previous month, 43.2 % retain their badges, while 56.8 % lose them in the current month.

In addition, 24.6 % of physicians have a pro bono identity each month. However, the average title level of physicians is below the chief physician or deputy chief physician. Nevertheless, most physicians work in higher-ranking hospitals, with an average cumulative number of 866 (logarithm value is 5.407) patients served per physician online and an average online consultation price of 36.37 (logarithm value is 3.235) renminbi. These model-free statistics offer profile information about physicians on the platform and allow us to better understand the factors that may influence patients' selecting physicians for consultations. We show the correlations between the variables in Table B2 in E-Companion B.

Some may worry that altruism could be another type of virtual badge, as it is captured by the existence of the pro bono identity of a physician. However, we would like to clarify that the pro bono identity is not a typical type of virtual badge, which is typically evaluated based

Table 1
Summary statistics and definitions of variables.

Variables	Definitions	Mean	SD	Min	Max
Log (NumCnslts$_{it}$)	Log number of online consultations performed by physician *i* in month *t*.	1.199	1.584	0	7.847
Badge	Dummy variable, 1 if physician *i* has a virtual badge in month *t* and 0 otherwise.	0.020	0.14	0	1
Badges_Win$_{it}$	Dummy variable, 1 if physician *i* has a virtual badge in month *t* but not in period *t-1* and 0 otherwise.	0.004	0.06	0	1
Badges_Keep$_{it}$	Dummy variable, 1 if physician *i* has a virtual badge in month *t* and also in period *t-1* and 0 otherwise.	0.013	0.111	0	1
Badges_Lose$_{it}$	Dummy variable, 1 if physician *i* has a virtual badge in month *t-1* but not in period *t* and 0 otherwise.	0.005	0.070	0	1
Altruism$_{it}$	Dummy variable, 1 if physician *i* has the pro bono identity in month *t* and 0 otherwise.	0.246	0.431	0	1
Title$_{it}$	Categorical variable, 2 if physician *i* is a chief physician in month *t*, 1 if physician *i* is a deputy chief physician in month *t*, and 0 for other titles.	0.579	0.722	0	2
Log(Price$_{it}$)	Log consultation price of physician *i* in month *t*	3.235	0.822	0.010	6.909
Hos_level$_{it}$	Dummy variable, equal to 1 if physician *i*'s hospital is at the highest level (i.e., Grade 3 Class A) in month t and 0 otherwise. The hospital levels, ranked from highest to lowest in China, are as follows: Grade 3 Class A, Grade 3 Class B, Grade 2 Class A, Grade 2 Class B, Grade 1 Class A, Grade 1 Class B.	0.715	0.451	0	1
Log (CumPatient$_{it}$)	Log number of cumulative patients treated by physician *i* in month *t*.	5.407	1.536	3.045	11.532
RatPosRvw$_{it}$	The ratio of the cumulative number of positive reviews to the cumulative number of consultations for physician *i* in month *t*	0.138	0.096	0	0.661

on user performance. For instance, in e-commerce platforms, whether merchants receive virtual badges is evaluated based on performance indicators such as feedback scores and repurchase rates [11]. In online healthcare platforms, virtual badges are typically awarded to physicians based on their performance, such as online ratings and response speeds [41]. Based on the existing literature, we consider that the pro bono identity in our study is not a typical type of virtual badge, and it is less commonly treated as a virtual badge in practice. Instead, it is more commonly served as a representation of altruism, which is based on whether a physician participates in voluntary activities or not. Moreover, our study focuses on the moderating effect of the pro bono identity, which represents the physicians' altruism, on the main relationship between virtual badges and patients' participation in online consultations, rather than studying the main effect of the pro bono identity as another type of virtual badges.

5. Empirical model and results

In this section, we examine the impact of temporarily acquired virtual badges on patients' subsequent participation in online consultations (i.e., consultation volume) to answer our first research question and provide empirical results (§5.1). We recognize that virtual badges may be assigned to physicians according to specific rules, which may lead to endogeneity concerns related to our results. Therefore, we employ alternative identification strategies and conduct several robustness checks to validate the main effect and mitigate endogeneity concerns (§5.2–5.3). Finally, we address the other research question, which explores the moderating roles of physicians' seniority and altruism in the main relationship (§5.4).

5.1. Impact of temporarily acquired virtual badges on patients' participation in online consultations

In this section, we examine the impact of temporarily acquired virtual badges on patients' participation in online consultations (i.e., consultation volume of physicians). Specifically, we first investigate whether the presence or absence of temporarily acquired virtual badges affects patients' participation in online consultation in the following fixed-effects model:

$$Y_{it} = \beta_0 + \beta_1 Badges_{it-1} + Controls + \alpha_i + \delta_t + \varepsilon_{it}, \quad (1)$$

where the dependent variable Y_{it} is *Log(NumCnslts$_{it}$ +1)*, which refers to the log number of online consultations performed by physician *i* in month *t*. We log-transform the *NumCnslts* variable to reduce skewness and produce elasticity-based explanations [64]. Since the variable may have a value of 0, we add 1 to it before the transformation. Regarding independent variables, *Badges$_{it-1}$* is a dummy variable indicating whether physician *i* has a badge at the beginning of month *t* (1 indicates physician *i* has a badge at the beginning of month *t*, otherwise 0). This status at the beginning of the month *t* is dependent on the physician's performance in the preceding month *t-1*. In essence, we investigate the effects of the badge's status at the start of the month on the subsequent consultation volume, rather than investigating whether a physician's performance in the current month influences their badge acquisition at the beginning of next month. In this fixed-effects model, we control the unobserved time-invariant physician-level characteristics, denoted as α_i. We also control the time effect, expressed as δ_t. Our coefficient of interest in Eq. (1) is β_1, which characterizes the difference between the consultation volume of physicians with virtual badges and those without badges.

Next, we examine the impact of dynamic changes in temporarily acquired virtual badges (i.e., physicians acquiring, retaining, and losing badges) on patients' subsequent participation in online consultations in the following fixed-effects model:

$$Y_{it} = \beta_0 + \beta_1 Badges_Win_{it-1} + \beta_2 Badges_Keep_{it-1} + \beta_3 Badges_Lose_{it-1} + Controls + \alpha_i + \delta_t + \varepsilon_{it}, \quad (2)$$

where $Badges_Win_{it-1}$, $Badges_Keep_{it-1}$, and $Badges_Lose_{it-1}$ are dummy variables indicating the dynamics of the physician i's badge in month t compared to month t-1. For example, if the physician i has no badge in month t-1 but has a badge in month t, $Badges_Win_{it-1}$ takes the value of 1, otherwise 0. If the physician i has a badge in both month t-1 and month t, $Badges_Keep_{it-1}$ takes the value of 1, otherwise 0. If the physician i has a badge in month t-1 but has no badge in month t, $Badges_Lose_{it-1}$ takes the value of 1, otherwise 0. For physicians who have a badge in both month t-1 and month t, or in month t-1 only, or in month t only, the values of the three binary variables $Badges_Win_{it-1}$, $Badges_Keep_{it-1}$, and $Badges_Lose_{it-1}$ for physician i in month t should add up to 1. In this fixed-effects model, we control the unobserved time-invariant physician-level characteristics, denoted as α_i. We also control the time effect, expressed as δ_t.

Our control variables (denoted as *Controls*) include the log price of the physician's consultation service ($Log(Price)_{it}$), the level of the physician's hospital (Hos_level_{it}), and the cumulative number of patients the physician has treated on the platform since registration ($Log(CumPatient)_{it}$). As noted in the previous literature, the previous variables may influence physicians' online consultation volume [16,49,65]. In addition, literature suggests that the behavior of existing customers (e.g., positive online reviews) can influence potential customers' choices of products or services [66]. Therefore, we control for the variable related to existing patients' evaluation behavior, namely, the ratio of the cumulative number of positive reviews to the cumulative number of consultations for physicians ($RatPosRvw_{it}$). We can identify the effect of temporarily acquired virtual badges by controlling for the previous variables. The estimation results of the regression Eq. (1) and (2) are presented in Table 2.

Column (1) in Table 2 is the result of Eq. (1), indicating the average effect of virtual badge presence. We find that the coefficient of $Badges_{it-1}$ is positive and statistically significant (0.196, $p < 0.01$). Specifically, virtual badges are associated with a 21.7 % (i.e., $e^{0.196}$ - 1) increase in consultation volume. This finding suggests that physicians having badges at the beginning of the month positively affects their consultation volume during that month. Moreover, column (2) is the result of Eq. (2), showing the impacts of virtual badge dynamics. In column (2), the coefficients of $Badges_Win_{it-1}$ and $Badges_Keep_{it-1}$ show a significant positive effect on the acquisition (0.149, $p < 0.05$) and retention of virtual badges (0.195, $p < 0.05$), respectively. Specifically, acquiring virtual badges is associated with a 16.1 % (i.e., $e^{0.149}$ - 1) increase in consultation volume, while retaining virtual badges has a larger effect on consultation volume, i.e., increasing by 21.5 % (i.e., $e^{0.195}$ - 1). The difference between these effects is statistically significant (Wald test, $p < 0.05$), indicating that the positive effect of badge retention on consultation volume is significantly greater than that of badge acquisition. At the same time, we do not observe a significant effect of $Badges_Lose_{it-1}$ on the consultation volume.

These findings answer our first research question and support H1, H2 (a), and H2(b), indicating that physicians acquiring and retaining virtual badges help patients identify physicians' performance and build trust, which results in more subsequent participation by physicians in online consultations. Furthermore, the positive effect of retaining badges is greater than that of acquiring badges because retaining badges is a more costly and observable signal and thus produces a larger signaling effect. However, we do not observe a significant effect of $Badges_Lose_{it-1}$ on the consultation volume, which rejects the H3. One possible explanation is that since it is not easy for physicians to acquire badges, the loss of badges may not erode patient trust in the short term. Such residual trust may mitigate the negative impact of badge loss. Considering this possibility, we further examine the impact of losing badges within a long-time frame to explore its significance. For brevity, we detail this analysis in the E-Companion and present the results in Table B3. In Table B3, we find that losing badges has a negative impact on consultation volume over the long term. This suggests that the effect of losing badges is not immediate but rather takes time to manifest.

In addition, we conduct analysis using the cumulative number of virtual badges received by physicians as an independent variable (see Table B4 in E-Companion for more details). The result in Table B4 shows that the coefficient of $Badges_Accumulate_{it}$ is positive and statistically significant (0.031, $p < 0.05$). The result suggests that the more virtual badges physicians receive, the stronger the signaling effects of the badges, which may facilitate a greater increase in the subsequent consultation volume. This is consistent with the result for retaining virtual badges in Table 2. Moreover, to address the issue of multicollinearity among the independent variables, we test the model using the variance inflation factor (VIF) method [49,67]. We present the results of this analysis in Table B5 in E-Companion B and find that the VIF related to the variable is far below the conventionally accepted threshold (i.e., VIF < 10 [68]), which indicates that multicollinearity is not severe in our model.

Table 2
The impact of virtual badges on consultation volume.

	(1)	(2)
Variables	*Consultation Volume*	*Consultation Volume*
Badges	**0.196*****	
	(4.01)	
Badges_Win		**0.149****
		(2.29)
Badges_Keep		**0.195****
		(2.23)
Badges_Lose		−0.007
		(−0.13)
Log(Price)	−0.329***	−0.325***
	(−4.58)	(−4.49)
Hos_level	0.935***	0.934***
	(4.32)	(4.28)
Log(CumPatient)	0.785***	0.771***
	(9.06)	(8.29)
RatPosRvw	−8.708***	−8.926***
	(−8.53)	(−8.50)
Constant	−1.398**	−1.310**
	(−2.30)	(−2.04)
Observations	36,960	36,735
Physician FE	YES	YES
Month dummies	YES	YES

Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2. Addressing endogeneity

In our research, the status of physicians' temporarily acquired virtual badges is exogenous to patients, thus the endogeneity issue may be less of a concern. Nonetheless, certain physician-side factors associated with the virtual badge status may correlate with physicians' consultation behaviors, including consultation volume. Therefore, we adopt various causal identification strategies to alleviate the endogeneity concern caused by physician-side factors. Specifically, following literature [69], we consider the following scenarios: (1) If the underlying selection process for badges is known, we can use a Heckman-type model to account for this selection process explicitly (see §5.2.1); (2) If the underlying selection process for badges is unknown and driven by observable characteristics, we can use matching methods combined with a fixed-effects model to address the endogeneity issue (see §5.2.2); or (3) If the underlying selection process for the badge is unknown and driven by unobservable time-variant factors, we adopt placebo tests to confirm our causal identification (see §5.2.3). The results of these identification strategies suggest that our findings are credible.

5.2.1. Heckman-type model

As mentioned earlier, temporarily acquired virtual badges are given

by OHPs to physicians who perform well, so the badge selection process can be considered known to some extent. For example, the platform might score physicians based on common evaluation criteria, such as past ratings, the number of patients served, and response speed, to decide whether to award a physician a badge. In this case, we use a Heckman-type model to directly illustrate the selection process, thus addressing the self-selection concerns caused by known selection processes [69]. We model whether a physician receives a badge by building a Probit model. The selection process is presented as follows:

$$\omega_{it}^{*} = \gamma_0 + \gamma_1 Rating_{it-1} + \gamma_2 Patient_{it-1} + \gamma_3 ResponseSpeed_{it-1} + \gamma_4 Controls + \varepsilon_{it}, \quad (3)$$

where ω_{it}^{*} is a latent variable, $Rating_{it-1}$ is the average score given by patients of physician i in month $t-1$, $Patient_{it-1}$ is the number of patients treated by physician i in month $t-1$, and $ResponseSpeed_{it-1}$ is the average response speed of physician i in month $t-1$ (divided into four levels; the higher the value, the faster the response speed). The indicator $Badges_{it-1} = 1$ if $\omega_{it}^{*} \geq 0$; $Badges_{it-1} = 0$ otherwise. Following Kumar et al. [69], we use a two-step procedure to estimate Eq. (1) and Eq. (3) jointly to address the endogeneity issue resulting from the known selection process.

Based on previous research, the inverse Mills ratio (IMR) should be included to control for unobserved factors when employing the Heckman-type model [70]. Specifically, the IMR coefficient reveals the presence of sample selection bias in the main estimation [71]. If the coefficient of IMR is significant, it indicates covariance between the error terms in the selection model and the outcome model, leading to estimation bias. Adding the IMR in the second stage (i.e., the outcome model) makes it possible to control for the selection bias calculated from the first stage [72]. The result in column (2) of Table 3 shows that the coefficient of the IMR in the second stage is significantly negative. This indicates that there is indeed a sample selection bias in the acquisition of virtual badges and that the consultation volume has been systematically underestimated. That is, without accounting for sample selection bias, the estimation of consultation volume would be biased downwards. Therefore, it is necessary to use the Heckman-type model to correct this bias.

After correcting the selection bias, the estimation results of column (2) shown in Table 3 are consistent with our main findings, which demonstrate the robustness of our results. Furthermore, we observe that the coefficients estimated in the second stage are larger than those in the regression Eq. (1). This might be due to the inclusion of residuals from the selection equation, which introduces additional information or more effectively controls for unobserved factors. In this case, the obtained coefficients are closer to the true causal effect.

Table 3
The impact of virtual badges on consultation volume: Heckman-type model.

	(1)	(2)
Variables	*Badges*	*Consultation Volume*
Badges		1.556***
		(6.62)
IMR		**−0.731*** **
		(−5.84)
l.Rating	1.393***	
	(4.74)	
l.Patient	0.523***	
	(18.52))	
l.ResponseSpeed	0.379***	
	(10.61)	
Constant	−16.13***	−0.330
	(−10.36)	(−0.40)
Observations	29,530	29,497
Controls	YES	YES
Physician dummies	No	YES
Month dummies	No	YES

Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2.2. Matching methods

Although the Heckman-type model can alleviate the endogeneity concern caused by the known underlying selection process, it assumes that Eq. (3) describes the real badge selection process. To relax this strong assumption and avoid model dependence, we combine a fixed-effects model with the propensity score matching (PSM) method to address unknown badge selection processes driven by observable features. The PSM method is a commonly used matching technique that enables randomized treatment by creating a similar control group for the treatment group in terms of observable characteristics [73,74]. However, PSM is biased in the case of selection-on-unobservable [75]. To address this issue, we use a fixed-effects model combined with PSM to eliminate bias caused by unobservable characteristics that do not vary over time.

In the matching process, we first randomly sort all the physicians to ensure that the sorting does not interfere with the matching process. Second, we perform logit regression to calculate propensity scores using a set of physicians' observable characteristics at the beginning of the sample period: (1) the average rating given by patients, (2) the log price of consultations, (3) the log cumulative number of patients treated on the platform, and (4) the average response speed. These variables reflect evaluations from patients [76,77] and physicians' online experience (such as cumulative number of patients served, price of consultations, and speed of response) [11], all of which can influence whether physicians receive a virtual badge and the volume of online consultations. Therefore, we select these physicians' characteristics as matching variables to obtain a more balanced sample.

Finally, we match the treated physicians who receive a badge to the most similar individuals in the control group based on propensity scores. Similar to Jung et al. [74], we use the one-to-one nearest neighbor matching method with replacement with a caliber of 0.05 to construct matched samples. Table 4 shows the summary statistics of the matching variables for the treatment and control groups before and after matching. We find no significant differences in observable variables between the matched treatment and control groups, implying that the two groups are well-balanced after matching.

In addition, to mitigate concerns about contingent results due to the selection of matching methods, we employ another matching method, coarsened exact matching (CEM). This method does not require checking covariate balance and has lower model dependence, so it is commonly used in the information systems literature [78]. We re-estimate Eq. (2) using the matched sample obtained from PSM and CEM matching methods and present results in Table 5. The results in Table 5 are qualitatively consistent with those in Table 2, proving the robustness of our findings.

5.2.3. Placebo test

Although the matching method and fixed-effects models can address biases caused by time-invariant observable and unobservable factors, they cannot yet address those endogeneity concerns caused by time-variant unobservable factors. Therefore, we follow Bertrand et al. [79] to conduct a placebo test to examine whether these factors would affect our results. We generate fictitious badges at each period in the placebo test and randomly assign them to some physicians. Since badges are fictitious, a significant effect of badges at the 5 % level should be found about 5 % of the time. If a significant effect is found at the 5 % level much >5 % of the time, it suggests that a placebo effect may drive our results. Otherwise, the results are not influenced by the unobservable variables.

Specifically, we generate fictitious badges and randomly assign them to a subset of physicians based on the proportion of the actual distribution of badges for the placebo test. Following Kumar et al. [67], we re-estimate regression Eq. (1) based on these placebo badges and repeat it 1000 times. Since these badges are fictitious, we expect the results to reject the null hypothesis of no effect in about 50 out of the 1000 runs. Our results show that for the dependent variable (i.e., consultation

Table 4
Summary statistics of treatment and control groups before and after matching.

Variables	Unmatched / Matched	Mean: Treated	Mean: Control	% bias	% reduce: Bias	*t*-test: t	*t*-test: $p > \|t\|$
Rating	Before	4.910	4.734	37.9	97.2	7.47	.000
	After	4.905	4.910	−1.1		−0.50	.620
ResponseSpeed	Before	3.579	2.725	104.7	99.4	23.65	.000
	After	3.492	3.497	−0.6		−0.13	.893
Log(Price)	Before	3.659	3.172	66.7	98.1	17.64	.000
	After	3.496	3.505	−1.3		−0.23	.816
Log(CumPatient)	Before	7.416	6.371	80.1	99.6	19.38	.000
	After	7.253	7.249	0.3		0.06	.951

Table 5
The impact of virtual badges on consultation volume with matched samples.

Variables	CEM		PSM	
	(1) Consultation Volume	*(2) Consultation Volume*	*(3) Consultation Volume*	*(4) Consultation Volume*
Badges	**0.194*****		**0.242*****	
	(3.98)		(4.52)	
Badges_Win		**0.156****		**0.202*****
		(2.46)		(2.96)
Badges_Keep		**0.203****		**0.295*****
		(2.32)		(3.10)
Badges_Lose		−0.002		0.033
		(−0.04)		(0.57)
Constant	0.108	0.836	2.993*	3.680**
	(0.09)	(0.59)	(1.82)	(2.15)
Observations	6039	5825	3734	3542
Controls	YES	YES	YES	YES
Physician FE	YES	YES	YES	YES
Month dummies	YES	YES	YES	YES

Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

volume), the fraction of simulations in which the null hypothesis is rejected is 4.7 % (47 times). The result demonstrates that our findings are unlikely to be influenced by a placebo effect.

5.3. Additional robustness checks

In this section, we report various robustness checks, the results of which are similar to the main findings reported in previous sections. The findings from these robustness checks suggest that our results are trustworthy and are not likely to be influenced by confounding factors.

5.3.1. Alternative measure for time window

To address potential concerns about the robustness of our results due to the time window of analysis, we rerun our main model using a different analysis time window. Specifically, we analyze a dataset using a six-month time window (i.e., August 2022 to January 2023) following the time window used in the main analysis to observe dynamic changes of temporarily acquired virtual badges on patients' subsequent participation in online consultations. The results in Table B6 in E-Companion B are consistent with our main findings in Table 2. Consequently, we ensure that our results are not influenced by the analysis time window, implying that patients increase participation in online consultations when physicians acquire or retain virtual badges.

5.3.2. Alternative identification method

Finally, we adopt a difference-in-difference (DID) model to estimate the overall effect of the temporarily acquired virtual badge. In previous sections, we examine the effect of badges on physicians at the individual level using a fixed-effects model. To further understand the impact of badges on the overall physician population, we conduct DID analyses. The DID model is widely used in natural experiment research designs to estimate the effect of an intervention or treatment by comparing outcomes over time between treatment and control groups [31,69]. In general, researchers need pre- and post-treatment data for both treatment and control groups to estimate the average effect caused by the external shock. However, our study window does not capture data prior to the introduction of the badge feature, so we group physicians and perform DID analysis based on their receipt of badges within the time window (i.e., after the badge feature is launched).

Specifically, we set physicians who receive badges at least once during the time window as the treatment group and consider them to have been consistently affected by the badge after receiving it. In contrast, physicians who never receive a badge are set as the control group and are considered unaffected by the badge. Therefore, we do not consider dynamic changes in the status of temporarily acquired virtual badges in this analysis. We combine the DID analysis with the matching method for regression to mitigate self-selection concerns caused by observable characteristics. In addition, we add physician and time-fixed effects to the model to control for unobservable time-invariant factors. We present the results in column (1) of Table B7 in E-Companion B. The findings are consistent with those in column (1) of Table 2, suggesting that the presence of virtual badges could improve online consultation volume across the overall physician population.

5.4. Moderating effects

Having estimated the main effects of temporarily acquired virtual badges on patients' participation in online consultations, we next answer our research question (2): how the effect of temporarily acquired virtual badges is moderated by physicians' seniority and altruism. Exploring the moderating roles in the main relationship allows us to understand which mechanisms are most prominent. Specifically, we add a moderating variable to Eq. (2) and interact with indicator variables representing badge status to determine its moderating role in the effect of temporarily acquired virtual badges.

5.4.1. The moderating role of physicians' seniority

First, we explore the moderating role of physicians' seniority in the impact of temporarily acquired virtual badges on patients' participation in online consultations. Following Zhang et al. [80], we measure the seniority of physicians by clinical title, which reflects their expertise and experience. In China, physicians' titles are determined through rigorous national examinations, and the government supervises the prerequisites and evaluation criteria for title evaluation [14]. Therefore, physicians' seniority can be validly measured by titles, and the interpretation of titles does not vary across hospitals. From higher to lower, the most common titles are chief physician, deputy chief physician, attending physician, and resident physician. Following previous literature (e.g.,

[81]), we categorize these titles into three levels to indicate physicians' seniority. Specifically, $Title_{it} = 2$ if the physician is a chief physician, $Title_{it} = 1$ if the physician is a deputy chief physician, and 0 in other cases. The regression equation is shown below.

$$Y_{it} = \beta_0 + \beta_1 Badges_Win_{it-1} + \beta_2 Badges_Keep_{it-1} + \beta_3 Badges_Lose_{it-1} + \beta_4 Badges_Win_{it-1} \times Title_{it} + \beta_5 Badges_Keep_{it-1} \times Title_{it} + \beta_6 Badges_Lose_{it-1} \times Title_{it} + Controls + \alpha_i + \delta_t + \varepsilon_{it}, \quad (4)$$

The estimation results are presented in column (1) of Table 6. The coefficients of the interaction terms ($Badges_Win_{it-1}$ and $Title_{it}$; $Badges_Keep_{it-1}$ and $Title_{it}$) are significant and negative (−0.254 and −0.323, respectively; $p < 0.1$), suggesting that the impact of the acquisition and retention of temporarily acquired virtual badges on the consultation volume is weaker for physicians with higher seniority. These findings support H4(a) and (b), indicating that high seniority indicates physicians' greater professional capability and experience, which leads to greater patient trust in senior physicians. Thus, the high seniority may diminish the positive effect of temporarily acquired virtual badges on consultation volume. In contrast, virtual badges can compensate for the lack of physicians' seniority for junior physicians, thus enhancing patients' trust in physicians and subsequently increasing their participation in online consultations. Therefore, the impact of temporarily acquired virtual badges on patients is smaller among senior physicians compared to junior physicians. However, we find an insignificant coefficient on the interaction term of $Badges_Lose_{it-1}$ and $Title_{it}$, which does not support H4(c). This suggests that patients may not perceive the loss of badges as a signal of a significant decline in physicians' performance in the short term, regardless of whether physicians are senior or junior.

5.4.2. The moderating role of physicians' altruism

We then investigate the interaction between the temporarily acquired virtual badges and physicians' altruism, an essential component of physician professionalism [59]. We use the pro bono identity granted to physicians by the OHP to capture their altruism. Pro bono identities are identity symbols granted by the platform to physicians who have engaged in a certain amount of prosocial activities on OHPs, such as providing free consultations or free health-related articles to patients. These prosocial activities reflect the physician's altruism, indicating the physician's caring for patient health and well-being beyond their self-interest [82]. Consequently, physicians with pro bono identities are generally perceived as more altruistic by patients compared to those without such identities. The pro bono identities are displayed on physicians' homepages, making them easily observed by patients. Moreover, during our observed time window, a substantial proportion of physicians (i.e., 49.6 %) has experienced changes in their pro bono identities. For notation, the variable $Altruism_{it}$ is denoted as 1 if physician i has the pro bono identity in month t, and 0 otherwise. Our regression model is as follows:

$$Y_{it} = \beta_0 + \beta_1 Badges_Win_{it-1} + \beta_2 Badges_Keep_{it-1} + \beta_3 Badges_Lose_{it-1} + \beta_4 Badges_Win_{it-1} \times Altruism_{it} + \beta_5 Badges_Keep_{it-1} \times Altruism_{it} + \beta_6 Badges_Lose_{it-1} \times Altruism_{it} + Controls + \alpha_i + \delta_t + \varepsilon_{it}, \quad (5)$$

We present the results of our moderation analysis by physicians' altruism in column (2) of Table 6. We find the coefficients on the interaction terms ($Badges_Win_{it-1}$ and $Altruism_{it}$; $Badges_Keep_{it-1}$ and $Altruism_{it}$) are significant and positive (0.207 and 0.098, respectively; $p < 0.1$), suggesting that patients are more likely to consult with physicians with higher levels of altruism than those with lower level of altruism. These findings support H5(a) and (b), indicating that physicians' altruism has a positive moderating effect on the relationship between the acquisition and retention of temporarily acquired virtual badges and the subsequent consultation volume. A possible reason for this result is that altruism serves as a signal of physicians' benevolence, and patients may build more trust in physicians with higher (vs. lower) levels of benevolence. This, in turn, facilitates patients to consult more with physicians with higher altruism. Meanwhile, we find an insignificant coefficient on the interaction term of $Badges_Lose_{it-1}$ and $Altruism_{it}$, indicating that the effect of the loss of badges on consultation volume does not differ significantly across groups of physicians with different levels of altruism, which does not support H5(c). One possible explanation is that the loss of badges is common among most physicians. As a result, patients do not interpret this loss as an indication of reduced performance and competence, and therefore, altruism does not exert a moderating effect.

6. Discussion and conclusion

The question of how non-financial, gamification design affects patients' subsequent participation in online consultations (i.e., consultation volume) has not yet been explored in the literature, and it is an important question since patients' online participation has a direct impact on the benefits and development of physicians and OHPs [16]. Importantly, there is scarce literature that studies the effect of virtual badges with a temporarily acquired nature, which is an increasingly popular type of badge design in practice [9].

In this study, we fill this research gap by investigating the signaling impact of the dynamic changes of temporarily acquired virtual badge status (i.e., acquisition, retention, and loss) on patients' participation in online consultations characterized by consultation volume. Our study contributes to the literature on patients' participation in OHPs, platform strategies, and signaling effects in online healthcare. In addition, our findings provide insights for practitioners and researchers on the effectiveness of virtual badges on OHPs and offer implications for the use and design of such non-financial incentives on online platforms.

Our empirical analyses reveal several key findings regarding the dynamic effects of temporarily acquired virtual badges on patients' subsequent participation in online consultations. First, we find that physicians acquiring virtual badges can improve patients' participation in consultations while retaining these badges further amplifies this positive effect. Our findings suggest that badge acquisition helps patients differentiate high-performance physicians from those with low performance, signaling to patients the previous performance of badge recipients. Such signals conveyed by acquiring virtual badges can enhance patients' trust in physicians and facilitate their consultation

Table 6
The moderating role of physicians' seniority and altruism.

	(1)	(2)
Variables	*Consultation Volume*	*Consultation Volume*
Badges_Win × *Title*	**−0.254****	
	(−2.04)	
Badges_Keep × *Title*	**−0.323***	
	(−1.92)	
Badges_Lose × *Title*	−0.085	
	(−0.78)	
Badges_Win × *Altruism*		**0.207***
		(1.87)
Badges_Keep × *Altruism*		**0.098****
		(2.08)
Badges_Lose × *Altruism*		0.081
		(0.97)
Badges_Win	0.256***	0.063
	(2.82)	(1.16)
Badges_Keep	0.334***	0.139*
	(2.61)	(1.68)
Badges_Lose	0.029	−0.050
	(0.32)	(−0.72)
Constant	−1.416**	−1.879***
	(−2.20)	(−2.90)
Observations	36,735	36,735
Controls	YES	YES
Physician FE	YES	YES
Month dummies	YES	YES

Robust t-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

decisions. Badge retention strengthens this positive effect by conveying more credible signals (in terms of signal cost and observability) to patients, which further enhances patients' trust and encourages their participation in online consultations. Understanding this dynamic is essential for encouraging and sustaining the engagement of patients and physicians on OHPs. Specifically, our study highlights the critical implications of badge retention from a signaling perspective in enhancing patients' trust and the effectiveness of temporarily acquired badges.

Second, we identify the distinct temporal dynamics in badge loss in terms of immediate and long-term effects in the context of OHPs. We find that the loss of badges does not immediately decrease consultation volume, suggesting that the temporary absence of badges does not negatively affect patients in the short term. In other words, even if physicians lose their badges, the positive effects of attaining badges can be restored if they regain the badges in a timely manner. On the other hand, our extended analysis of the long-term effects of losing badges shows that the negative impact of losing badges on consultation volume takes time to manifest. This may lead to a reduction in both the consultation volume and patients' engagement in online consultations in the long run. This finding underscores the importance of studying temporarily acquired badges because it indicates that badge loss is not without consequence but has enduring negative effects over a relatively long term, which also suggests that such badge losses may potentially demotivate physicians' contributions. Overall, the insights on the loss of badges highlight the unique effects of temporarily acquired virtual badges in online healthcare.

Third, we gauge the moderating roles of physicians' key characteristics (i.e., seniority and altruism) in the relationship between the dynamic changes of temporarily acquired virtual badges and patients' subsequent participation in online consultations. Specifically, we observe a negative moderating role of physicians' seniority, suggesting that acquiring and retaining virtual badges have stronger effects on increasing the consultation volume for junior physicians than for senior physicians. In addition, our research shows that physicians' altruism positively moderates the dynamic effects of temporarily acquired virtual badges. This suggests that altruism can work with virtual badges to enhance patients' trust, thus exerting a positive moderating effect on consultation volume. Our findings on the heterogeneous impacts of virtual badges across physicians' characteristics provide insights for managers of OHPs on the effectiveness of temporarily acquired virtual badge design on consultation outcomes for different physicians.

6.1. Managerial implications

Our answers to the first research question suggest that the non-financial, gamification design of OHPs (i.e., temporarily acquired virtual badges) plays an important role in enhancing patients' participation in online medical consultations. Specifically, the temporarily acquired design of badges can capture a physician's current performance, which may increase patients' trust and attract more patients to consult with physicians. These findings suggest that physicians acquiring and retaining temporarily acquired badges on OHPs can improve patients' participation in online consultations. Our study provides practical insights for OHP managers that OHPs can further leverage the utility of temporarily acquired badges by enhancing their visibility, such as providing information on the cost and effort required for physicians to acquire the badge. These designs can further strengthen the credibility of the badges and improve patients' trust in physicians, thereby increasing the consultation volume on OHPs.

In addition, we find that the loss of badges does not immediately yield a negative impact but takes time to manifest. This result suggests that badge loss can reduce patients' participation over the long term, which may also demotivate physicians from providing consultations on OHPs. Our findings provide practical implications that platform managers need to be aware of the negative influence of badge loss over the long term on patients' participation and physicians' motivation in providing consultations. Accordingly, platform managers may consider implementing other incentives (e.g., small financial rewards) to encourage physicians to provide quality consultations when their badges are revoked.

We also emphasize the generalizability of our findings on temporarily acquired virtual badges on OHPs. On the one hand, like our focal OHP that implemented the temporarily acquired virtual badges, several other OHPs have also adopted similar temporarily acquired gamification designs, such as Umer Doctor, which provides a weekly ranking of physicians. Therefore, our practical implications are not limited to a specific platform but can be extended to other OHPs. On the other hand, the presence of temporarily acquired virtual badges is essentially a signaling strategy, and our findings on such badges can be extended to other online two-sided platforms from a signaling perspective. Specifically, temporarily acquired badges dynamically reflect physicians' recent performance and convey a signal of recency attribute to patients (i.e., up-to-date assessment of physicians' capabilities), thus influencing patients' subsequent participation in online consultations. On other online platforms utilizing virtual badges, the temporarily acquired design of virtual badges may also affect users through a signaling effect on facilitating user demand for the provider's service.

To answer the second research question, we further examine the moderating roles of physicians' key characteristics (i.e., seniority and altruism). Our findings provide guidance that can help platform managers understand the differential effects of temporarily acquired virtual badges across physicians and utilize flexible gamification design. First, our results on the negative moderating effect of physicians' seniority on the consultation volume suggest that physicians with higher seniority are less responsive to temporarily acquired virtual badges. This finding reminds managers of OHPs that temporarily acquired badges that show the recognition of competency are not very effective for senior physicians. Therefore, managers of OHPs should focus more on non-senior physicians when designing and optimizing temporarily acquired badges. Second, our findings about the positive moderating effect of physicians' altruism suggest that the positive dynamic effect of temporarily acquired badges is more pronounced among physicians with higher levels of altruism, as altruism further enhances patients' trust in these physicians. Overall, our findings regarding the moderating role of physicians' characteristics provide practical insights for the design and

implementation of virtual badges on OHPs.

Our findings on the heterogeneous effects of physicians' altruism and seniority in moderating the effects of temporarily acquired virtual badges on consultation volume can be generalized to similar online healthcare environments in other countries. Specifically, in some countries (e.g., the United States, India), online healthcare platforms (e.g., MDLIVE, Practo) allow physicians to show their seniority and altruism status. For example, on Practo, physicians can show their altruism by answering patients' health questions for free (e.g., "Consult Q&A" service) and can publicize their seniority by showing years of clinical experience on their online profiles. Therefore, our findings regarding the moderating effects of physicians' altruism and seniority can be generalized to similar OHPs in other countries.

6.2. Theoretical implications

First, our study provides insights into the impact of platform strategies (i.e., virtual badges) and patients' participation in OHPs. Research has focused on the effect of virtual badges on users' online behavior. However, most of them emphasize the effects of virtual badges evaluated by individuals' cumulative performance [39,40]. For example, Airbnb awards hosts with "Superhost" badges based on their cumulative performance over a year, such as the number of bookings and response rate [40]. However, research on virtual badges with temporary designs is scarce, and their effectiveness on patients is unknown. Our study fills this gap by investigating the impact of monthly dynamic changes of temporarily acquired virtual badges (i.e., acquisition, retention, and loss) on patients' participation in online consultations. We provide empirical evidence that acquiring and retaining temporarily acquired virtual badges can improve patients' participation in online consultations.

Second, we enrich signaling theory in online healthcare by exploring the signaling effects of recency attribute to patients (i.e., up-to-date assessment of physicians' performance via temporarily acquired virtual badges). Relevant literature has applied the theory to OHPs to explain the effects of physician signals on patients [22,44,45]. For example, Chen et al. [44] use signaling theory to explain the effectiveness of two signals (i.e., earned and owned signals) on patient–physician matching on OHPs. However, previous research has primarily focused on the impact of signals of cumulative attributes (i.e., evaluations based on physicians' historical behaviors) on patients. Our study contributes to signaling effects research by exploring the impact of recency–attribute signals conveyed by temporarily acquired badges on patients' participation in online consultations, which has not been systematically studied.

Third, our study complements the literature on the heterogeneous effects of platform strategies by identifying the moderating roles of physicians' key characteristics (i.e., seniority and altruism) on the effect of temporarily acquired virtual badges. Previous related research explored the moderating role of physicians' seniority in the effect of platform strategies (e.g., online–offline service integration) on patients [65] and the role of altruism in the effect of financial incentives on online reviewers' contribution behavior [83]. Our study contributes to this stream of literature by examining the moderating role of physicians' seniority and altruism in the relationship between temporarily acquired virtual badges and patients' subsequent participation in online consultation.

6.3. Limitations and future research

Our study is subject to a few limitations, but it also provides opportunities for future research. First, we use observational data combined with signaling theory to explain the relationship between virtual badges and patients' subsequent online participation. To validate our findings, future research could use survey data to quantify the underlying process of patients' motivation. Second, we use multiple identification strategies to account for possible self-selection problems due to the non-random assignment of virtual badges, and the endogeneity concerns that this assignment poses. Nevertheless, we acknowledge that there may still be endogeneity concerns associated with our results, and these concerns could be addressed in the future using randomized field experiments.

CRediT authorship contribution statement

Yunting Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ran (Alan) Zhang:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization. **Min Zhang:** Writing – review & editing, Software, Formal analysis. **Xitong Guo:** Validation, Supervision, Funding acquisition.

Declaration of competing interest

There is no financial or personal conflict of interest.

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Supplementary materials

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References

[1] J. Cordina, J. Fowkes, R. Malani, L. Medford-Davis, Patients love telehealth–physicians are not so sure. https://www.mckinsey.com/industries/healthcare-systems-and-services/our-insights/patients-love-telehealth-physicians-are-not-so-sure, 2022.
[2] R. Pearl, B. Wayling, The telehealth era is just beginning. https://hbr.org/2022/05/the-telehealth-era-is-just-beginning, 2022.
[3] D.H. Saifee, Z. (Eric) Zheng, I.R. Bardhan, A. Lahiri, Are online reviews of physicians reliable indicators of clinical outcomes? A focus on chronic disease management, Inf. Syst. Res. 31 (2020) 1282–1300.
[4] J. Liu, X. Zhang, F. Meng, K. Lai, Deploying gamification to engage physicians in an online health community: an operational paradox, Int. J. Prod. Econ. 228 (2020) 107847.
[5] Even28, Dentist "badges,". https://even28.com/dentist-badges-dentist-page/, 2022.
[6] J.R. Fanfarelli, P.M. Harden, It's badging time! permanence, expiration, and lifespan of digital badges, in: 16th Int. Conf. Found. Digit. Games 2021, , ACM, New York, NY, USA, 2021, pp. 1–10.
[7] R. McDaniel, J. Fanfarelli, Building better digital badges: pairing completion logic with psychological factors, Simul. Gaming 47 (2016) 73–102.
[8] A. Schmidt, What's the hype with digital badges?. https://valueinsights.ch/whats-the-hype-with-digital-badges/, 2020.
[9] Credly, Can digital badges be fake? Understanding the risks and solutions. https://learn.credly.com/blog/can-digital-badges-be-fake-understanding-the-risks-and-solutions, 2024.
[10] B.L. Connelly, S.T. Certo, R.D. Ireland, C.R. Reutzel, Signaling theory: a review and assessment, J. Manage. 37 (2011) 39–67.
[11] H.K. Cheng, W. Fan, P. Guo, H. Huang, L. Qiu, Can "Gold Medal" online sellers earn gold? The impact of reputation badges on sales, J. Manag. Inf. Syst. 37 (2020) 1099–1127.
[12] J. Hamari, J. Koivisto, Working out for likes": an empirical study on social influence in exercise gamification, Comput. Human Behav. 50 (2015) 333–347.
[13] L. (Dorothy) Jiang, J. (Jove) Hou, X. Ma, P.A. Pavlou, Punished for success? A natural experiment of displaying clinical hospital quality on review platforms, Inf. Syst. Res. 36 (2025) 285–306.
[14] S. Guo, X. Guo, Y. Fang, D. Vogel, How doctors gain social and economic returns in online health-care communities: a professional capital perspective, J. Manag. Inf. Syst. 34 (2017) 487–519.
[15] M. Pandya, How does a google trust badge help you win online customers' trust?. https://customerthink.com/how-does-a-google-trust-badge-help-you-win-online-customers-trust/, 2022.

[16] Q.Ben Liu, X. Liu, X. Guo, The effects of participating in a physician-driven online health community in managing chronic disease: evidence from two natural experiments, MIS Q. 44 (2020) 391–419.
[17] L. Wang, L.L. Yan, T. Zhou, X. Guo, G.R. Heim, Understanding physicians' online-offline behavior dynamics: an empirical study, Inf. Syst. Res. 31 (2020) 537–555.
[18] N. Dhingra, J. Emmett, A. Samo, B. Schaninger, Igniting individual purpose in times of crisis. https://www.mckinsey.com/capabilities/people-and-organizational-performance/our-insights/igniting-individual-purpose-in-times-of-crisis, 2020.
[19] A. Bhattacherjee, Individual trust in online firms: scale development and initial test, J. Manag. Inf. Syst. 19 (2002) 211–241.
[20] R. Aggarwal, V. Midha, N. Sullivan, Superlatives and scope of improvement in online recommendations: breath of life or a kiss of death? MIS Q. 45 (2021) 1411–1432.
[21] J. Harris, Altruism: should it be included as an attribute of medical professionalism? Heal. Prof. Educ. 4 (2018) 3–8.
[22] J. Zhou, R. Kishore, L. Amo, C. Ye, Description and demonstration signals as complements and substitutes in an online market for mental health care, MIS Q. 46 (2022) 2055–2084.
[23] H. Cavusoglu, Z. Li, S.H. Kim, How do virtual badges incentivize voluntary contributions to online communities? Inf. Manag. 58 (2021) 103483.
[24] S. Liu, H. Wang, B. Gao, Z. Deng, Doctors' provision of online health consultation service and patient review valence: evidence from a quasi-experiment, Inf. Manag. 59 (2022) 103360.
[25] M. Barrett, E. Oborn, W. Orlikowski, Creating value in online communities: the sociomaterial configuring of strategy, platform, and stakeholder engagement, Inf. Syst. Res. 27 (2016) 704–723.
[26] Y. Liu, C. Ren, D. Shi, K. Li, X. Zhang, Evaluating the social value of online health information for third-party patients: is uncertainty always bad? Inf. Process. Manag. 57 (2020) 102259.
[27] S. Chen, X. Guo, T. Wu, X. Ju, Exploring the online doctor-patient interaction on patient satisfaction based on text mining and empirical analysis, Inf. Process. Manag. 57 (2020) 102253.
[28] T. Mirzaei, P. Esmaeilzadeh, Engagement in online health communities: channel expansion and social exchanges, Inf. Manag. 58 (2021) 103404.
[29] A.S. Navathe, K.G. Volpp, K.L. Caldarella, A. Bond, A.B. Troxel, J. Zhu, S. Matloubieh, Z. Lyon, A.Mishra Meza, L. Sacks, C. Nelson, P. Patel, J. Shea, D. Calcagno, S. Vittore, K. Sokol, K. Weng, N. McDowald, P. Crawford, D. Small, E. J. Emanuel, Effect of financial bonus size, loss aversion, and increased social pressure on physician pay-for-performance, JAMA Netw. Open 2 (2019) e187950.
[30] X. Liu, X. Guo, H. Wu, T. Wu, The impact of individual and organizational reputation on physicians' appointments online, Int. J. Electron. Commer. 20 (2016) 551–577.
[31] S. Khurana, L. Qiu, S. Kumar, When a doctor knows, it shows: an empirical analysis of doctors' responses in a Q&A forum of an online healthcare portal, Inf. Syst. Res. 30 (2019) 872–891.
[32] X. Zhang, W. Chen, B. Gu, X. Guo, The impact of introductory incentives on new users : evidence from an online health community, in: ICIS 2017 Proceeding, 2017, pp. 1–19.
[33] H. Wang, R. Du, W. Shen, L. Qiu, W. Fan, Product reviews: a benefit, a burden, or a trifle? How seller reputation affects the role of product reviews, MIS Q. 46 (2022) 1243–1272.
[34] D. Ko, F. Mai, Z. Shan, D. Zhang, Operational efficiency and patient-centered health care: a view from online physician reviews, J. Oper. Manag. 65 (2019) 353–379.
[35] Y. Xu, M. Armony, A. Ghose, The interplay between online reviews and physician demand: an empirical investigation, Manage. Sci. 67 (2021) 7344–7361.
[36] J. Kim, A. Gambino, Do we trust the crowd or information system? Effects of personalization and bandwagon cues on users' attitudes and behavioral intentions toward a restaurant recommendation website, Comput. Human Behav. 65 (2016) 369–379.
[37] L. Khansa, X. Ma, D. Liginlal, S.S. Kim, Understanding members' active participation in online question-and-answer communities: a theory and empirical analysis, J. Manag. Inf. Syst. 32 (2015) 162–203.
[38] S. Bhattacharyya, S. Banerjee, I. Bose, One size does not fit all: rethinking recognition system design for behaviorally heterogeneous online communities, Inf. Manag. 57 (2020) 103245.
[39] Y.R. Tausczik, J.W. Pennebaker, Predicting the perceived quality of online mathematics contributions from users' reputations, Conf. Hum. Factors Comput. Syst. (2011) 1885–1888.
[40] S. Liang, M. Schuckert, R. Law, C.C. Chen, Be a "Superhost": the importance of badge systems for peer-to-peer rental accommodations, Tour. Manag. 60 (2017) 454–465.
[41] P. Ouyang, J.J. Wang, U. Ali, The impact of gamification on the patient's engagement in the online health community, Aslib J. Inf. Manag. 74 (2022) 1196–1213.
[42] L. Zhu, G. Yin, W. He, Is this opinion leader's review useful? Peripheral cues for online review helpfulness, J. Electron. Commer. Res. 15 (2014) 267–280.
[43] W. Wang, J. Luo, M. Dugas, G. (Gordon) Gao, R. Agarwal, R.M. Werner, Recency of online physician ratings, JAMA Intern. Med. 182 (2022) 881.
[44] L. Chen, A. Rai, W. Chen, X. Guo, Signaling effects under dynamic capacity in online matching platforms: evidence from online health consultation communities, Inf. Syst. Res. 36 (2025) 202–224.
[45] M. Spence, Job market signaling. Uncertain. Econ., Elsevier, 1978, pp. 281–306.
[46] S. He, B. Hollenbeck, D. Proserpio, The market for fake reviews, Mark. Sci. 41 (2022) 896–921.
[47] M. Kokkodis, T. Lappas, G.C. Kane, Optional purchase verification in e-commerce platforms: more representative product ratings and higher quality reviews, Prod. Oper. Manag. 31 (2022) 2943–2961.
[48] W.E. Baker, N. Bulkley, Paying it forward vs. rewarding reputation: mechanisms of generalized reciprocity, Organ. Sci. 25 (2014) 1493–1510.
[49] W. Fan, Q. Zhou, L. Qiu, S. Kumar, Should doctors open online consultation services? An empirical investigation of their impact on offline appointments, Inf. Syst. Res. 34 (2023) 629–651.
[50] K. Dopelt, Y.G. Bachner, J. Urkin, Z. Yahav, N. Davidovitch, P. Barach, Perceptions of practicing physicians and members of the public on the attributes of a "Good Doctor, Healthcare 10 (2021) 73.
[51] M. Hojat, D.Z. Louis, K. Maxwell, F. Markham, R. Wender, J.S. Gonnella, Patient perceptions of physician empathy, satisfaction with physician, interpersonal trust, and compliance, Int. J. Med. Educ. 1 (2010) 83–87.
[52] L. Wang, K. Gunasti, R. Shankar, J. Pancras, R. Gopal, Impact of gamification on perceptions of word-of-mouth contributors and actions of word-of-mouth consumers, MIS Q. 44 (2020) 1987–2011.
[53] M. Spence, Job market signaling, Q. J. Econ. 87 (1973) 355–374.
[54] M. Lin, Y. Liu, S. Viswanathan, Effectiveness of reputation in contracting for customized production: evidence from online labor markets, Manage. Sci. 64 (2018) 345–359.
[55] X. Zhang, X. Guo, K. hung Lai, W. Yi, How does online interactional unfairness matter for patient–doctor relationship quality in online health consultation? The contingencies of professional seniority and disease severity, Eur. J. Inf. Syst. 28 (2019) 336–354.
[56] X. Tian, W. Li, B. Zhou, B. Zhang, X. Wang, Y. Yin, J. Yu, X. Wu, X. Sun, C. Meng, Influence of physicians' seniority on efficacy of silver needle therapy for low back fasciitis: a prospective, cohort study, Medicine 102 (2023) e33862.
[57] S. Liu, M. Zhang, B. Gao, G. Jiang, Physician voice characteristics and patient satisfaction in online health consultation, Inf. Manag. 57 (2020) 103233.
[58] O. Arazy, N. Kumar, B. Shapira, A theory-driven design framework for social recommender systems, J. Assoc. Inf. Syst. 11 (2010) 455–490.
[59] J. Coulehan, P.C. Williams, Vanquishing virtue: the impact of medical education, Acad. Med. 76 (2001) 598–605.
[60] S.R. Cruess, R.L. Cruess, Professionalism must be taught, BMJ 315 (1997) 1674–1677.
[61] D.H. McKnight, V. Choudhury, C. Kacmar, Developing and validating trust measures for e-commerce: an integrative typology, Inf. Syst. Res. 13 (2002) 334–359.
[62] N. Nguyen, A. Leclerc, The effect of service employees' competence on financial institutions' image: benevolence as a moderator variable, J. Serv. Mark. 25 (2011) 349–360.
[63] H. Xuan, Ali Health financial report reveals key breakthroughs as internet healthcare finds new possibilities. https://www.vbdata.cn/55147, 2020.
[64] W. Zhao, Q. Ben Liu, X. Guo, T. Wu, S. Kumar, Quid pro quo in online medical consultation? Investigating the effects of small monetary gifts from patients, Prod. Oper. Manag. 31 (2022) 1698–1718.
[65] N. Huang, Z. Yan, H. Yin, Effects of online–offline service integration on e-healthcare providers: a quasi-natural experiment, Prod. Oper. Manag. 30 (2021) 2359–2378.
[66] I.E. Vermeulen, D. Seegers, Tried and tested: the impact of online hotel reviews on consumer consideration, Tour. Manag. 30 (2009) 123–127.
[67] N. Kumar, L. Qiu, S. Kumar, A hashtag is worth a thousand words: an empirical investigation of social media strategies in trademarking hashtags, Inf. Syst. Res. 33 (2022) 1403–1427.
[68] S. Chatterjee, A.S. Hadi, Regression Analysis by Example, 5th Edit, John Wiley & Sons, Hoboken, New Jersey, 2006.
[69] N. Kumar, L. Qiu, S. Kumar, Exit, voice, and response on digital platforms: an empirical investigation of online management response strategies, Inf. Syst. Res. 29 (2018) 849–870.
[70] H.A. Lee, A.A. Choi, T. Sun, W. Oh, Reviewing before reading? An empirical investigation of book-consumption patterns and their effects on reviews and sales, Inf. Syst. Res. 32 (2021) 1368–1389.
[71] D. Salerno, G. Sampagnaro, V. Verdoliva, Fintech and IPO underpricing: an explorative study, Financ. Res. Lett. 44 (2022) 102071.
[72] M.H. Tajvarpour, D. Pujari, Bigger from a distance: the moderating role of spatial distance on the importance of traditional and rhetorical quality signals for transactions in crowdfunding, Decis. Support Syst. 156 (2022) 113742.
[73] L. Kuang, N. Huang, Y. Hong, Z. Yan, Spillover effects of financial incentives on non-incentivized user engagement: evidence from an online knowledge exchange platform, J. Manag. Inf. Syst. 36 (2019) 289–320.
[74] J. Jung, R. Bapna, J. Ramaprasad, A. Umyarov, Love unshackled: identifying the effect of mobile app adoption in online dating, MIS Q. 43 (2019) 47–72.
[75] S. Mithas, M.S. Krishnan, From association to causation via a potential outcomes approach, Inf. Syst. Res. 20 (2009) 295–313.
[76] T.L. Botelho, M. Gertsberg, The disciplining effect of status: evaluator status awards and observed gender bias in evaluations, Manage. Sci. 68 (2022) 5311–5329.
[77] Y. Huang, P. Vir Singh, K. Srinivasan, Crowdsourcing new product ideas under consumer learning, Manage. Sci. 60 (2014) 2138–2159.
[78] S. Eftekhari, N. Yaraghi, R.D. Gopal, R. Ramesh, Impact of health information exchange adoption on referral patterns, Manage. Sci. 69 (2023) 1615–1638.
[79] M. Bertrand, E. Duflo, S. Mullainathan, How much should we trust differences-in-differences estimates? Q. J. Econ. 119 (2004) 249–275.
[80] X. Zhang, F. Guo, T. Xu, Y. Li, What motivates physicians to share free health information on online health platforms ? Inf. Process. Manag. 57 (2020) 102166.

[81] Y. Liu, X. Xu, Y. Jin, H. Deng, Understanding the digital resilience of physicians during the COVID-19 pandemic: an empirical study, MIS Q. 47 (2023) 391–422.
[82] J. Li, W.H. Dow, S. Kariv, Social preferences of future physicians, Proc. Natl. Acad. Sci. U.S.A. 114 (2017) E10291–E10300.
[83] D. Qiao, S.Y. Lee, A.B. Whinston, Q. Wei, Financial incentives dampen altruism in online prosocial contributions: a study of online reviews, Inf. Syst. Res. 31 (2020) 1361–1375.

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